جامعة التقنية والعلوم التطبيقية بـإبـراء

University of Technology and Applied Sciences - Ibra

DEPARTMENT OF INFORMATION TECHNOLOGY

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|---|---|---|
| **Academic year: 2021-2022** | **Semester: 3** | **Level: Diploma I** |
| **Course Code: ITSE101** | **Course Name: Programming I** | **Section: 1** |
| **Student ID:----------** | **Student Name:-------** | |

## I. DEFINE AND ANALYZE THE PROBLEM (2 marks)

**Problem description:**

Choose your home carefully. We write program for Future Home company in Muscat. Our company will calculator a month house rent. The program will read your name, number, age, status, and months do you want to rent the house. There are four types of houses: Detached House, Flat, Villa, and Bungalow. The cost of type of house Detached House, Flat, Villa, Bungalow = (70,50,100,60) OMR respectively. User will choose type of his \ her house. Also, user can choose house with view or not. If user choose view, user should choose Beach, buildings, and Forest. The cost of view Beach, buildings, Forest = (40,20,30) OMR respectively. After that, user will choose type of neighborhood. If user choose calm, the cost is 20 OMR. If user choose noisy, the cost is 5 OMR. Then, user will choose House Furniture. If user want furniture, the cost is 90 OMR. Else the cost is 0 OMR. The cost of house is calculator type of house, view, type of neighborhood and house furniture. The cost of house for many months that user will choose is the cost of house multiple months. The program will display the cost of house and Payment method. If user choose Monthly Rent, the cost of house divides months or can choose cash. Display the house is ready and thank user.

1. **Aim:** To find a home quickly, easily, reliably, and of your choice.

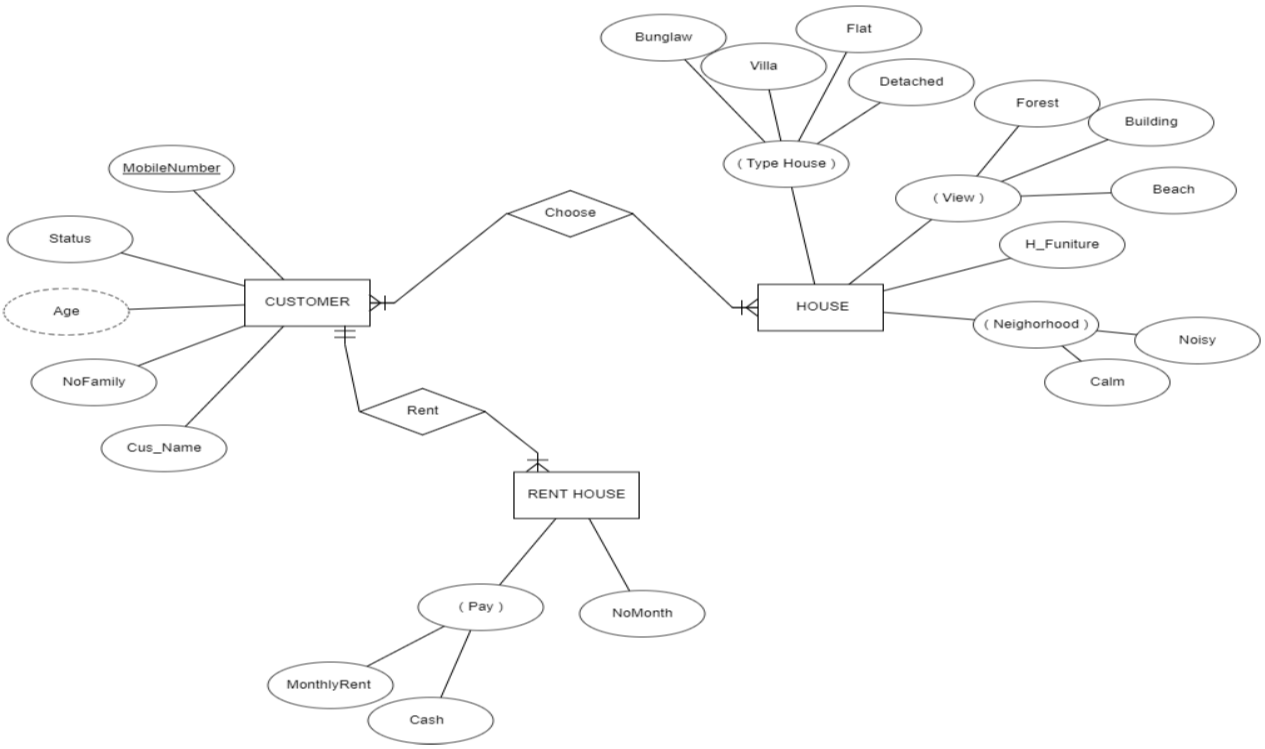
2. **Data that will be used in the program with their type:**
   - Number of Months : Integer
   - Type and view of house : Will choose number for your choice (Integer)
   - User Name and statues : String
   - Type of Neighborhood : Will choose number for your choice (Integer)
   - House Furniture : Will choose number for your choice (Integer)
   - The Pay : Will choose number for your choice (Integer)

3. **Inputs Required:**
   - Name
   - Age
   - Mobile Number
   - Status
   - Number of member family
   - Number of Months

4. **Process/Calculation:**
   - Total = cost of type house+ Cost of View+ Cost of Type of Neighborhood + House Furniture
   - Total for All Months= Total*Number of Months
5. **Output to be produced:**
   - The cost of house
   - Monthly rent
   - The Pay Methods
   - House Summary

## II. DESIGNING (1 mark)

**Algorithm/Flowchart/Class diagram/ER Diagram (as applicable)**




## III. CODING

a. **Which iteration is most suitable to be used in solving this problem? Justify your answer (0.5 mark)**

- CONDITIONAL EXECUTION
- ALTERNATIVE EXECUTION
- CHAINED CONDITIONALS
- NESTED CONDITIONALS
- WHILE LOOP

**b. Program:** **(6 marks)**

In [8]:

```
ans="y"
while ans=="y" or ans=="Y":
    n=str(input("Enter Your Name  :  "))
    a=int(input("Enter Your Age    :  "))
    s=str(input("Enter Your Status   : "))
    b=int(input("How Many Members are In The Family : "))
    c=int(input("Enter your Mobile Number : "))
    z=int(input("How Many Months do You Want to Rent The Houes : "))
    print("                                                                ")
    print("-----------*************-------------***********-----------------**********-----------")
    print("                                    Type of house  ")
    print("                                                                ")
    print("1- Detached House         2- Flat             3- Villa               4- Bunglaw")
    h=int(input("Enter type of house : "))
    if h==3:
        m=(100*z)
        aa="Villa"
        print(f"The cost of Villa : {m} OMR")
    elif h==1:
        m=(70*z)
        aa="Detached House"
        print(f"The cost of Detached House : {m} OMR")
    elif h==2:
        m=(50*z)
        aa="Flat"
        print(f"The cost of Flat : {m} OMR")
    elif h==4:
        m=(60*z)
        aa="Bunglaw"
        print(f"The cost of Bunglaw : {m} OMR")
    else:
        print("You should Choose the type of house from 1 to 4")
    print("-----------*************-------------***********-----------------**********-----------")
    print("                                    View for Your house ")
    print("1- Yes                                                      2- No")
    q=int(input("Do you want view for your new house : "))
    if q==1:
        print("1- Beach                        2-buildings                        3-Forest")
        w=int(input("A number for your choice : "))
        if w==1:
            e=40
            cc="Beach"
            print(f"The cost of your house with Beach : {e} OMR")
        elif w==2:
            e=20
            cc= "Buildings"
            print(f"The cost of your house with Buildings : {e} OMR")
        elif w==3:
            e=30
            cc="Forest"
            print(f"The cost of your house with Forest : {e} OMR")
        else:
            print("Please, Choose Your house view")
    else:
        print("Your home without view ")
        e=0
    print("-----------*************-------------***********-----------------**********-----------")
    print("                                    Type of neighborhood")
    print(" 1- calm                                                      2-noisy")
    o=int(input("Enter a number for your neighborhood : "))
    if o==1:
        g=20
        bb= "Your house in calm neighborhood"
        print(f"Your house in calm neighborhood, so the cost is {g} OMR")
    elif o==2:
        g=5
        bb= "Your house in noisy neighborhood "
        print(f"Your house in noisy neighborhood , so the cost is  {g} OMR")
    else:
        print("Your choice is not avilable ")
    print("-----------*************-------------***********-----------------**********-----------")
    print("                                    Your House Furniture   ")
    print("1- Furniture                                                 2- Without Furniture")
    v=int(input("Enter a Number for your choice : "))
    if v==1:
        l=90
        dd="It has furniture"
        print(f"The Cost of Your House with Furniture : {l} OMR")
    elif v==2:
        l=0
        dd="It is without furniture"
        print("The Cost of Your House without Furniture")
    else:
        print("Please, Choose Your house Furniture")
    print("-----------*************-------------***********-----------------**********-----------")
    n=(m+g+l+e)
    print(f"Your House Cost is {n} OMR")
    print("-----------*************-------------***********-----------------**********-----------")
    print("                                    The Pay")
    print("1- Monthly Rent                                              2-Cash            ")
    x=int(input("Enter Your Payment method : "))
    if x==1:
        qq=(n/z)
        print (f"You should pay  {qq} monthely")
    elif x==2:
        print("Please Pay in advance ")
    else:
        print("Please Your Payment method ")
    print("-----------*************-------------***********-----------------**********-----------")
```

```
    print("----------**********-----------**********----------**********----------")
    print("                                        Your House Summary")
    print(f"Your {aa}")
    if q==1:
        print(f"It has {cc} View ")
    else:
        print("It dose not have Veiw ")
    if o==1:
        print(bb)
    else:
        print(bb)
    if v==1:
        print(dd)
    else:
        print(dd)
    print("----------**********-----------**********----------**********----------")
    print("                                Thank You. Your House is ready.")
    ans=str(input("Do you want do agine?  if yes write 'y' or 'Y' otherwise No write 'N' : "))
    if ans=="Y" or ans=="y":
        continue
    else:
        break
```

```
Enter Your Name  :  ZUMURUD
Enter Your Age    :  19
Enter Your Status   : Unmarried
How Many Members are In The Family : 3
Enter your Mobile Number : 9999999999
How Many Months do You Want to Rent The Houes : 12

----------**********----------**********----------**********----------
                                Type of house

1- Detached House         2- Flat             3- Villa                4- Bunglaw
Enter type of house : 3
The cost of Villa : 1200 OMR
----------**********----------**********----------**********----------
                               View for Your house
1- Yes                                                  2- No
Do you want view for your new house : 1
1- Beach                        2-buildings                     3-Forest
A number for your choice : 3
The cost of your house with Forest : 30 OMR
----------**********----------**********----------**********----------
                               Type of neighborhood
 1- calm                                                  2-noisy
Enter a number for your neighborhood : 1
Your house in calm neighborhood, so the cost is 20 OMR
----------**********----------**********----------**********----------
                                Your House Furniture
1- Furniture                                            2- Without Furniture
Enter a Number for your choice : 1
The Cost of Your House with Furniture : 90 OMR
----------**********----------**********----------**********----------
Your House Cost is 1340 OMR
----------**********----------**********----------**********----------
                                   The Pay
1- Monthly Rent                                          2-Cash
Enter Your Payment method : 2
Please Pay in advance
----------**********----------**********----------**********----------
                                   Your House Summary
Your Villa
It has Forest View
Your house in calm neighborhood
It has furniture
----------**********----------**********----------**********----------
                               Thank You. Your House is ready.
Do you want do agine?  if yes write 'y' or 'Y' otherwise No write 'N' : Y
Enter Your Name  :  SHEFA
Enter Your Age    :  20
Enter Your Status   : Unmarried
How Many Members are In The Family : 3
Enter your Mobile Number : 99999999
How Many Months do You Want to Rent The Houes : 13

----------**********----------**********----------**********----------
                                Type of house

1- Detached House         2- Flat             3- Villa                4- Bunglaw
Enter type of house : 2
The cost of Flat : 650 OMR
----------**********----------**********----------**********----------
                               View for Your house
1- Yes                                                  2- No
Do you want view for your new house : 1
1- Beach                        2-buildings                     3-Forest
A number for your choice : 1
The cost of your house with Beach : 40 OMR
----------**********----------**********----------**********----------
                               Type of neighborhood
```

```
                                Type of neighborhood
 1- calm                                                    2-noisy
Enter a number for your neighborhood : 1
Your house in calm neighborhood, so the cost is 20 OMR
----------**************-------------**********-----------------**********-----------
                                Your House Furniture
1- Furniture                                          2- Without Furniture
Enter a Number for your choice : 1
The Cost of Your House with Furniture : 90 OMR
----------**************-------------**********-----------------**********-----------
Your House Cost is 800 OMR
----------**************-------------**********-----------------**********-----------
                                    The Pay
1- Monthly Rent                                       2-Cash
Enter Your Payment method : 1
You should pay  61.53846153846154 monthely
  ----------**************-------------**********-----------------**********-----------
                                    Your House Summary
  Your Flat
  It has Beach View
  Your house in calm neighborhood
  It has furniture
  ----------**************-------------**********-----------------**********-----------
                                Thank You. Your House is ready.
  Do you want do agine?  if yes write 'y' or 'Y' otherwise No write 'N' : N
```

## IV. TESTING AND DEBUGGING

1. **The Errors you have encountered and how did you debug them** (list a maximum of 5 errors only) **(0.5 mark)**

```
1- Detached House         2- Flat             3- Villa                4- Bunglaw
Enter type of house : 3
The cost of Villa : 900 OMR
----------**************-------------**********-----------------**********-----------
                                View for Your house
1- Yes                                                  2- No
Do you want view for your new house : 2

---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Input In [12], in <cell line: 35>()
     33 print("1- Yes                                                  2- No")
     34 Q=int(input("Do you want view for your new house : "))
---> 35 if q==1:
     36     print("1- Beach                        2-buildings                     3-Forest")
     37     w=int(input("A number for your choice : "))

NameError: name 'q' is not defined
```

```
if v==1:
    print(dd)
else:
    print(dd)
print("----------**************-------------**********-----------------**********-----------")
print("                                Thank You. Your House is ready.")
```

```
  File <tokenize>:42
    elif w==2:
    ^
IndentationError: unindent does not match any outer indentation level
```

```
        print(dd)
    else:
        print(dd)
print(f"Your {aa}. {ll}. {ff}. {ww}")
print("----------**************-------------**********-----------------**********-----------")
print("                                Thank You. Your House is ready.")
```

```
  File <tokenize>:101
    else:
    ^
IndentationError: unindent does not match any outer indentation level
```

```
In [ ]:
```

```
1- Furniture                                                    2- Without Furniture
Enter a Number for your choice : 1
The Cost of Your House with Furniture : 90 OMR
----------**************-------------***********-----------------**********-----------

-------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Input In [17], in <cell line: 84>()
     82     print("Please, Choose Your house Furniture")
     83 print("----------**************-------------***********-----------------**********-----------")
---> 84 n=(m+g+l+e)
     85 print(f"Your House Cost is {n} OMR")
     86 print("----------**************-------------***********-----------------**********-----------")

NameError: name 'g' is not defined
```

```
    print("It dose not have Veiw ")
if o==1:
    print(bb)
else:
    print(bb)
if v==1:
    print(dd)
else:
    print(dd)
print("----------**************-------------***********-----------------**********-----------")
print("                                Thank You. Your House is ready.")
```

```
  Input In [11]
    else h==4:
         ^
SyntaxError: invalid syntax
```

2. **Sample Output**: (Output should be displayed for all types of conditions. Minimum of two outputs) **(1 mark)**

- **CHAINED CONDITIONALS**

```
----------**************-------------***********-----------------**********-----------
                                Type of neighborhood
 1- calm                                                        2-noise
Enter a number for your neighborhood : 1
Your house in calm neighborhood, so the cost is 20 OMR
----------**************-------------***********-----------------**********-----------
```

- **NESTED CONDITIONALS**

```
    print("It dose not have Veiw ")
if o==1:
    print(bb)
else:
    print(bb)
if v==1:
    print(dd)
else:
    print(dd)
print("----------**************-------------***********-----------------**********-----------")
print("                                Thank You. Your House is ready.")
```

```
  Input In [10]
    print(f"The Cost of Your House with Furniture :",1 "OMR")
                                                      ^
SyntaxError: invalid syntax
```

```
----------**************----------**********-----------------**********----------
                                View for Your house
1- Yes                                                      2- No
Do you want view for your new house : 1
1- Beach                    2-buildings                         3-Forest
A number for your choice : 1
The cost of your house with Beach : 40 OMR
----------**************----------**********-----------------**********----------
```

- **CONDITIONAL EXECUTION**

```
----------**************----------**********-----------------**********----------
                                Your House Summary
Your Villa
It has Beach View
Your house in calm neighborhood
It has furniture
----------**************----------**********-----------------**********----------
```

**REFERENCES(IF ANY) :** **(1 mark)**

(List the references you have used to understand and solve the problem: Class material/Reference books with Author, Publisher name & Publishing date/Webpage URL with the date accessed)

Note: Use the APA style of referencing as shown:

Citing a book: Author, A. (Year of Publication). *Title of work*. Publisher City, State: Publisher.

For example:

Finney, J. (1970). *Time and again*. New York, NY: Simon and Schuster.

Citing an e-book: Author, A. (Year of Publication). Title of work [E-Reader Version]. Retrieved from https://xxxx or DOI:xxxx

For Example:

Eggers, D. (2008). The circle [Kindle Version]. Retrieved from https://www.amazon.com/

*Reference:*
*APA Book Citation: retrieved from http://www.bibme.org/citation-guide/apa/book/*

| Sl. No. | Details | Purpose |
|---|---|---|
| 1 | Dive, M. (2009). House choice tips. New House. /https://dealapp.sa/blog/the-right-home-space | Get information about houses |
| 2 | Streefkerk, R. (2019, October 11). The most notable changes. Home. https://www.exaqr.com/advantages-disadvantages-buying- | House emergencies |
| 3 | Rdwan, N. (2001). Features of a house. Rent House. https://www.homify.com.eg/ideabooks/1193956/6- | .A future home |